

# Rohit Nagendra Prasad

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## EDUCATION

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Vellore Institute of Technology, Chennai

Bachelors in Computer Science with spl. in AI & ML; GPA: 9.01

Chennai, India

August 2023 - Present

Shrishti Vidyashram Sr. Sec. School

Till 12th grade; Percentage: 83.2

Vellore, India

June 2016 - April 2023

## SKILLS SUMMARY

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- **Languages:** Python, Kotlin, Java, C#, C++, SQL, JavaScript, TypeScript
- **Frameworks:** Spring Boot, Spring Data JPA, React, Vite, Flask, LangChain, RAG, Pandas, NumPy, Scikit-learn
- **Databases:** PostgreSQL, MySQL, OracleSQL
- **Tools:** Git, Docker, Maven, Flyway, Postman
- **Concepts:** REST APIs, JWT Auth, RBAC, System Design, API Integration, Workflow Automation, Async Processing

## WORK EXPERIENCE

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Software Engineering Intern | American Express | Bengaluru, India

May 2026 - July 2026

*ReleaseOps | Enterprise Release Observability Platform*

- Built an enterprise release operations platform integrating **6 enterprise systems**, replacing fragmented multi-tool release management with a centralized, traceable workflow for release planning, approvals, and operational visibility.
- Automated release governance by developing **40+ REST APIs** with secure role-based workflows, scheduled synchronization, and audit logging, significantly reducing manual effort required to manage production releases.
- Engineered **asynchronous synchronization pipelines** and **release conflict detection**, improving release traceability and accelerating cross-functional release reviews across enterprise systems.
- Integrated **AI-assisted release intelligence** with prompt injection safeguards to generate contextual release summaries, accelerating stakeholder reviews without compromising governance.

Product Development Intern | Medialogic Solutions Pvt. Ltd. | Chennai, India

June 2025 - July 2025

*SonoCare Reporting Platform | 4,000+ Hospitals Served*

- Built and deployed a production-grade speech-to-text system using **Google STT** and **Python.NET**, enabling live voice-driven data entry and voice-controlled tab navigation that reduced report completion time by **80%+**.
- Engineered structured field parsing and low-latency mapping logic for **C# WinForms**, streamlining diagnostic reporting workflows across **4,000+ hospitals**.
- Achieved **90%+ transcription accuracy** while reducing clinical terminology errors from **30% to ~5%** through RapidFuzz-based fuzzy matching and domain-specific medical parsing.

## PROJECTS

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EM-Connect | [GITHUB](#) | [LIVE](#)

February 2026 - Present

*Distributed Event Management Platform*

- Built a distributed event management platform comprising a **Spring Boot API**, **3 Go microservices**, **RabbitMQ**, and **PostgreSQL**, deployed as **4 independent cloud services** with asynchronous event-driven communication.
- Engineered concurrency-safe registration workflows using pessimistic locking, transactional guarantees, and asynchronous ticket generation, preventing duplicate registrations while maintaining data consistency under parallel requests.
- Designed and deployed an event-driven messaging topology with RabbitMQ topic exchanges, dead-letter queues, retry handling, and horizontally decoupled Go workers for scalable background processing.
- Implemented a real-time event broadcasting system using **WebSockets** and topic-based subscriptions, enabling live participant updates, event announcements, and resilient client auto-reconnection.
- Deployed the platform across **Netlify** and **Render** with Docker Compose-based local orchestration, integrating health monitoring and production-ready infrastructure for end-to-end cloud deployment.

- Built an AI-powered Quality Assurance assistant that integrates classical quality engineering tools with real-time AI guidance to support defect analysis and process improvement, targeting MSMEs and shop-floor users.
- Implemented interactive quality tools including Pareto charts, control charts, histograms, Cp/Cpk analysis, and fishbone diagrams, enabling data-driven root cause analysis from uploaded CSV and Excel datasets.
- Developed a full-stack web application using React, TypeScript, Tailwind CSS, and FastAPI, with real-time chart generation, export capabilities (PNG, PDF), and responsive UI.
- Applied statistical analysis and visualization pipelines using Pandas, NumPy, Matplotlib, Plotly, and Seaborn to automate quality insights and actionable recommendations.
- Project presented at the IFQM Conclave for Academia 2025 and awarded 1st place by Mr. Vivek Chaand Sehgal for innovation in AI-assisted quality management.

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## CERTIFICATIONS

**Oracle Cloud Infrastructure 2025 Certified Generative AI Professional | [CERTIFICATE](#)****July 2025**

- Earned OCI 2025 Generative AI Professional certification (exam 1Z0-1127-25), demonstrating expertise in large language model architecture, Retrieval-Augmented Generation (RAG), LangChain, and OCI Generative AI agent workflows.
- Acquired hands-on mastery of OCI Generative AI services including model endpoint creation, fine-tuning, and secure deployment atop Oracle's enterprise-grade infrastructure.

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## PUBLICATIONS

**Deep Learning Models for Intelligent IoT Ecosystems | [IGI Global](#)****February 2026***Book Chapter in AI in Advanced Systems, Robotics, and Healthcare*

- Co-authored a 36-page research chapter exploring the integration of deep learning with Internet of Things (IoT) ecosystems to enable intelligent device networks and automated decision systems.
- Reviewed major deep learning architectures including CNNs, RNNs, LSTMs, and autoencoders for large-scale sensor data processing and environmental monitoring applications.
- Analyzed emerging AIoT trends including lightweight edge models, federated learning, explainable AI, and next-generation connectivity through 5G/6G networks.

**ENDCL: Attention-Enhanced CNN-BiLSTM Model for Cardiovascular Disease Detection | [IEEE](#)****September 2025***2025 5th International Conference on Emerging Research in Electronics, Computer Science and Technology*

- Proposed a hybrid deep learning architecture combining CNN residual blocks, Bidirectional LSTM networks, and multi-head attention for automated cardiovascular disease detection.
- Trained and evaluated the model on the Cleveland Heart Disease dataset using SMOTE class balancing and feature normalization techniques.
- Achieved approximately 90% predictive accuracy with improved diagnostic reliability compared to traditional machine learning approaches.

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## PATENTS

**DETECTION OF UNDERGROUND WATER PIPELINES LEAKAGE USING AERIAL ACOUSTIC SYSTEM**Patent Application No: [202641016585](#)**Published February 20, 2026**

- Developed a UAV-based acoustic monitoring system for real-time detection of underground water pipeline leakages using aerial survey techniques.
- Designed a distributed network of passive surface acoustic wave sensors that convert pipe-wall vibrations caused by leaks into backscattered RF signals for remote sensing.

**SMART REFRIGERATION SYSTEM ENABLING OCCLUSION DETECTION AND TARGETED DEFROSTING, AND METHOD THEREOF**Patent Application No: [202541096122](#)**Published October 31, 2025**

- Designed a smart refrigeration system leveraging multimodal sensors and AI to detect occluded food items, generate a 3D occupancy map, and enable targeted, zone-specific defrosting only where required.
- Implemented intelligent compartment repositioning, cooling optimization, and synchronization of defrost cycles with user cooking schedules to improve energy efficiency and food preservation.